

About Innovation Pilots

The Innovation Pilots team works with Governments, Industry and the community, as well as partnering through the Catalysing Australia Biosecurity (CAB) initiative, to test new technologies and approaches to support a strong national biosecurity system, prepare for future challenges and help us to work smarter.

Our approach

We work across the department and partner with stakeholders to investigate innovative technologies that may improve how we work or address biosecurity risks. We test our ideas in short, sharp pilots within our operations and provide our evaluation and recommendations to the relevant business areas.

We have a strong network across government, research institutions and leading innovators and can connect you with the right people to bring your idea to life.

How we can help

If you have a business problem, creative idea or innovative solution for the department, the Innovation Pilots' team wants to know about it.

We welcome ideas from within our department, other government departments or private enterprises that have novel approaches, new technologies or innovations that can help us with our business challenges and enhance Australia's biosecurity.

To explore how we can work with you to find a solution or make your idea a reality, get in touch with the team at innovationpilots@aff.gov.au

Our projects

The Innovation Pilots team are leading around 50 projects in collaboration with department business areas, and also with external Australian and international parties, such as independent innovators, research and educational institutions, and research and innovation corporations.

We have completed 15 innovation pilots and have more than 30 pilots underway or being scoped.

Completed projects

Since commencing in 2018, the Innovation Pilots initiative has completed 15 pilots. Here are some examples of past pilots.

Spot the robot dog

In 2022, Spot's capabilities were trialled in several activities that involved inspecting used machinery, vehicles and containers for pests and other biosecurity risk material (BRM) such as soil, leaves and insects. The Innovation Pilot team has continued to explore Spot's capabilities, including the potential for him to undertake an automated internal, and external inspection of shipping containers to identify BRM. Initial trials were promising, but the exploration of the use of robotics and machine learning by the department is still in its infancy. The department plans to continue to explore possibilities through trials using Spot and other technologies.

Watch a video of a pilot project with the Boston Dynamics robotic dog Spot.

Introduction

This is the accessible text transcript of a video of Spot the robot dog designed by the American engineering and robotics design company Boston dynamics. Spot is performing various biosecurity activities.

Transcript

This is Spot. Also known as Boston Dynamic's robotic canine unit B59.

It may lack the smooth coat and pedigree of a real-life detector dog, but its capabilities extend beyond biosecurity.

Spot can carry up to 14kg of additional equipment. It has 360-degree perception.

Its smart sensors can fetch and bring back different types of data. Take images of biosecurity risks and detect harmful gases.

Spot is part of DAFF's fabulous Innovation Pilots Program which aims to explore and test robotic capabilities to see if technology can help us make tasks safer and more efficient.

Help staff perform biosecurity container inspections and collect environmental data including radiation readings. Essentially, determine if Spot can walk the walk.

The robotic canine was put through its paces. It was sent into, and under, heavy cargo containers at the Port of Darwin. Its finds were received by biosecurity officers from a safe distance, using tablets to upload Spot's data.

We also sent it out to an ex-Ranger uranium mine. Fitted with a gamma sensor, it conducted radionuclide counts over a mine rehabilitation site.

We ran it over tough terrain, in the hot temperatures and tested its 3D mapping ability.

So, did Spot pass muster? With a few improvements it could be a feasible solution for the department, and a robotic canine of our own is on its way to assist us in future pilot activities.

Take a bow Spot!

Ballast water analysis and sampling

Collecting samples of large organisms is challenging due to the large volume of water needed. A pilot was conducted to test the effectiveness of 2 water sampling kits to analyse ballast water as per international standards. Results found that both kits were useful in testing ballast water. The kits will soon be adopted by the department for future evaluation of ballast water compliance.

Hades-5Z remote controlled vehicles

As robot technology advances, more devices are being introduced that can help with day-to-day operations. The Hades-5Z robot was tested during various border inspections in Australia. With custom changes, this technology could be used to inspect hard-to-reach or high-risk areas.

Exciting projects underway

We currently have approximately 30 pilots underway or being scoped. You can read about some of them below.

Varroa mite detection technology

We're leading 2 pilot studies in collaboration with several government departments and education institutions. The studies will trial Vimana Tech's BeeRight technology which is a low-cost Internet of Things solution which allows beekeepers to remotely monitor and manage their hives. The pilot study will run in the Australian Capital Territory and New South Wales. The data collected will be used to determine if the technology has the potential to detect the presence of varroa mite.

Ethyl formate for treatment of khapra beetle

We're collaborating with Murdoch University to test the efficacy of ethyl formate as a potential treatment for khapra beetle. Laboratory research and field trials are taking place in India using live beetle specimens. If successful, it could be a different way to treat khapra beetle in sea containers, plant products and grain storages effectively, instead of using methyl bromide.

Aerial drone eDNA sampling

To assist officers in collecting environmental DNA samples from remote or difficult terrain that may be infested with invasive species such as deer, we're piloting the use of a HydraSleeve water sampler. This sampler is fixed to a commercially available drone and can collect water samples from separate water bodies in the Australian Capital Territory (ACT). The samples will be tested by the University of Canberra to provide accurate information on the presence or absence of non-native deer in the ACT.

Biosecurity Matters newsletter

Have you subscribed to Biosecurity Matters? This quarterly newsletter gives you an overview of the department's activities in managing biosecurity risks at the border and within Australia, including some of our exciting Innovation Pilot projects.

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Contact us

We're piloting new ideas now to optimise how we work today.

To share your idea or ask questions contact us at innovationpilots@aff.gov.au

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